Observables for Two-Qubit Quantum Sensors with Spin and Spatial-Mode Entanglement

Local, Nonlocal, and Hybrid Measurements; Metrology via (Q)FI; Connections to High-Energy Physics

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Executive summary (what you will learn and use)

We study a **two-qubit quantum sensor prototype** where two subsystems A, B each carry:

- a **spin qubit** (internal two-level system) and
- a **spatial-mode qubit** (two-mode/path/position subspace),

on an **unspecified experimental platform**.

This mini-course provides:

- A concrete **parameter-encoding** model (common-mode vs differential/gradient fields; spin, mode, and spin–mode couplings).
- A structured set of **observable families** for diagnosing entanglement and sensing performance: local, correlators, collective J and J^2, rotated parity, Bell/CHSH, and hybrid spin–spatial observables.
- A metrological bridge from observables to sensitivity using **Cramér–Rao bounds**, **Fisher information**, and **QFI** (including $F_Q = 4(\Delta G)^2$ for pure probes). (**???**)
- **Measurement circuits** (Quantikz) for correlators, parity (direct and ancilla), and CHSH settings, with compile-safe labels (math mode subscripts; `quantikz` not inside display math).
- Worked examples (Bell, singlet/triplet, spatial Bell, common vs differential sensitivity with explicit QFI tables)

Outline

- Physical setup and sensing generators (spin, spatial mode, hybrid).
- Metrological framework: FI, QFI, Cramér–Rao; generator variances.
- Observable families (definitions, what they probe, measurement circuits).
- Worked examples: Bell states; singlet/triplet; spatial two-mode Bell; common vs differential fields.
- Experimental considerations: dephasing, SPAM, tomography, shot budgets.
- HEP connections and operator mappings.
- Minimal measurement program + suggested experiments.
- Python simulation (NumPy) for probes, encoding, dephasing, FI/QFI.

Physical setup: two subsystems, spin + spatial-mode qubits

Platform: unspecified. Interpret A and B as two addressable sensors (ions, atoms, NVs, superconducting qubits, etc.).

Each subsystem $X \in \{A, B\}$ contains:

- **Spin qubit** Hilbert space $\mathcal{H}_{s,X} \cong \mathbb{C}^2$ with Pauli operators $\sigma_x^{(X)}, \sigma_y^{(X)}, \sigma_z^{(X)}$.
- **Spatial-mode qubit** Hilbert space $\mathcal{H}_{m,X} \cong \mathbb{C}^2$ with effective Pauli operators $\tau_x^{(X)}, \tau_y^{(X)}, \tau_z^{(X)}$ (two paths/modes/locations).

Total Hilbert space:

$$\mathcal{H} = (\mathcal{H}_{s,A} \otimes \mathcal{H}_{m,A}) \otimes (\mathcal{H}_{s,B} \otimes \mathcal{H}_{m,B}) \cong (\mathbb{C}^2)^{\otimes 4}.$$

Entanglement structure:

- Spin entanglement: across $\mathcal{H}_{s,A} \otimes \mathcal{H}_{s,B}$.
- Mode entanglement: across $\mathcal{H}_{m,A} \otimes \mathcal{H}_{m,B}$.
- Hybrid entanglement: involving spin and mode jointly (e.g., spin conditioned on mode).

Parameter encoding: unitary model and generators

A sensing experiment encodes a parameter ϕ via

$$\rho(\phi) = U(\phi)\,\rho_0\,U^\dagger(\phi), \qquad U(\phi) = e^{-i\phi G},$$

where G is the **generator** of the encoding. (**???**)

We distinguish three broad sensing scenarios:

1. **Spin-only encoding:** G built from σ operators.
2. **Mode-only encoding:** G built from τ operators (e.g., path potential differences).
3. **Hybrid encoding:** G couples spin and mode (e.g., position-dependent Zeeman shifts).

Key theme:

Metrological relevance depends on $(\Delta G)^2$ for the chosen G.

Entanglement helps if it increases the variance of the *right* generator.

Common-mode vs differential (gradient) spin fields

A minimal spin-field interaction:

$$H = \frac{\gamma}{2}\Big(B_A\sigma_z^{(A)} + B_B\sigma_z^{(B)}\Big).$$

Decompose

$$B_A = B + \frac{\delta B}{2}, \qquad B_B = B - \frac{\delta B}{2}.$$

Then for evolution time t:

$$U(t) = e^{-itH} = e^{-i\phi_{\mathrm{com}}\, G_{\mathrm{com}}^{(s)}}\, e^{-i\phi_{\mathrm{diff}}\, G_{\mathrm{diff}}^{(s)}},$$

with

$$G_{\mathrm{com}}^{(s)} = \frac{1}{2}(\sigma_z^{(A)} + \sigma_z^{(B)}), \quad G_{\mathrm{diff}}^{(s)} = \frac{1}{2}(\sigma_z^{(A)} - \sigma_z^{(B)}), \quad \phi_{\mathrm{com}} = \gamma B t, \quad \phi_{\mathrm{diff}} = \gamma\frac{\delta B}{2}t.$$

Different entangled probes optimize different tasks:

- $|\Phi\rangle$-type (e.g., $|\Phi^+\rangle$) are sensitive to common-mode $G_{\mathrm{com}}^{(s)}$.
- $|\Psi\rangle$-type (e.g., $|\Psi^+\rangle$) are sensitive to differential $G_{\mathrm{diff}}^{(s)}$.

Mode-phase encoding (two-mode spatial qubit)

Let each mode qubit have basis $|L\rangle$ and $|R\rangle$, with

$$\tau_z |L\rangle = +|L\rangle\,, \qquad \tau_z |R\rangle = -|R\rangle\,.$$

Then a spatial phase difference (e.g., potential imbalance between L and R) can be modeled as

$$H_m = \frac{\kappa}{2}\Big(V_A \tau_z^{(A)} + V_B \tau_z^{(B)}\Big) \quad \Rightarrow \quad U_m(t) = e^{-i\phi_m G_m}.$$

Define

$$G_{\mathrm{com}}^{(m)} = \frac{1}{2}(\tau_z^{(A)} + \tau_z^{(B)}), \qquad G_{\mathrm{diff}}^{(m)} = \frac{1}{2}(\tau_z^{(A)} - \tau_z^{(B)})$$

as common-mode and differential mode generators.

Parallelism to spin: spatial Bell states can act as gradiometers for differential phases.

Hybrid spin–mode encoding (position-dependent spin coupling)

If the field depends on spatial mode (e.g., opposite Zeeman shifts on two paths), a natural generator is

$$G_{sz} = \frac{1}{2}\left(\sigma_z^{(A)}\tau_z^{(A)} + \sigma_z^{(B)}\tau_z^{(B)}\right).$$

This encodes a parameter ϕ via

$$U(\phi) = \exp\left[-i\phi G_{sz}\right].$$

Consequences:

- Spin-only measurements can be insufficient: the signal is stored in spin *conditioned on mode*.
- Hybrid observables like $\sigma_z^{(A)}\tau_z^{(A)}$ and $(\sigma_z^{(A)}\tau_z^{(A)})(\sigma_z^{(B)}\tau_z^{(B)})$ become the relevant diagnostics.

Classical Fisher information and Cramér–Rao bound

Given measurement outcomes o with probabilities $p(o|\phi)$, the **classical Fisher information (FI)** is

$$F_C(\phi) = \sum_o \frac{(\partial_\phi p(o|\phi))^2}{p(o|\phi)}.$$

For ν independent repetitions (shots), the Cramér–Rao bound:

$$\mathrm{Var}(\hat{\phi}) \geq \frac{1}{\nu\, F_C(\phi)}.$$

This is the operational link between **chosen observable/measurement** and achievable sensitivity. (**?**)

Binary outcome example ($o \in \{+, -\}$):

$$F_C(\phi) = \frac{(\partial_\phi p_+)^2}{p_+} + \frac{(\partial_\phi p_-)^2}{p_-}, \quad p_- = 1 - p_+.$$

Quantum Fisher information (QFI) and ultimate precision

The **quantum Fisher information (QFI)** is the maximum FI over all POVMs:

$$F_Q(\phi) = \max_{\text{measurements}} F_C(\phi).$$

For unitary encoding $\rho(\phi) = e^{-i\phi G}\rho_0 e^{i\phi G}$, F_Q depends on ρ_0 and G. (**??**)

Pure probes $\rho_0 = |\psi_0\rangle\langle\psi_0|$:

$$\boxed{F_Q = 4(\Delta G)^2_{\psi_0}}.$$

Thus, generator variance is the central metrological resource. (**??**)

Mixed probes: the variance formula does not hold; one can use the spectral formula (next slide).

QFI for mixed states under unitary encoding (spectral formula)

Let $\rho_0 = \sum_i \lambda_i |i\rangle \langle i|$ (eigendecomposition), and the encoding is generated by Hermitian G. Then the QFI can be written as (**??**)

$$F_Q(\rho_0, G) = 2 \sum_{i,j:\, \lambda_i + \lambda_j > 0} \frac{(\lambda_i - \lambda_j)^2}{\lambda_i + \lambda_j} |\langle i| G |j\rangle|^2.$$

For a pure state ($\lambda_0 = 1$), this reduces to $F_Q = 4(\Delta G)^2$.

Practical use: this formula is what we will implement in the Python simulation for dephased (mixed) states.

Entanglement and QFI benchmarks for linear generators

For N qubits and a collective linear generator, separable states obey QFI bounds. A common statement: for a linear two-mode interferometer, separable probes satisfy $F_Q \leq N$. (**??**)

For **two qubits** ($N = 2$) and generators such as $G^{(s)}_{\mathrm{com}}$ or $G^{(s)}_{\mathrm{diff}}$:

$$F_Q \leq 2 \quad \text{for separable probes,} \qquad F_Q \leq 4 \quad \text{(two-qubit “Heisenberg” maximum).}$$

Thus $F_Q > 2$ certifies entanglement that is useful for estimating the parameter encoded by that generator.

Caveat: Entanglement is *task-dependent*. A state can be maximally entangled yet have $F_Q = 0$ for some generators.

Observable taxonomy (spin, mode, hybrid)

We use six observable families (spin forms shown; mode analogs via $\sigma \to \tau$):

1. **Local:** $\sigma_\alpha^{(A)}$, $\sigma_\alpha^{(B)}$.
2. **Two-body correlators:** $\sigma_\alpha^{(A)}\sigma_\beta^{(B)}$.
3. **Collective:** $J_\alpha = \frac{1}{2}(\sigma_\alpha^{(A)} + \sigma_\alpha^{(B)})$ and moments J_α^2.
4. **Parity (rotated parity):** $\Pi_z^{(s)} = \sigma_z^{(A)}\sigma_z^{(B)}$ and $R^\dagger \Pi_z^{(s)} R$.
5. **Bell/CHSH:** $\mathcal{B}_{\mathrm{CHSH}}$ built from correlators in rotated bases. (**?**)
6. **Hybrid spin–mode:** $\sigma_\alpha^{(A)}\tau_\mu^{(A)}$ and $(\sigma_\alpha^{(A)}\tau_\mu^{(A)})(\sigma_\beta^{(B)}\tau_\nu^{(B)})$.

Metrology link: Many relevant generators are *collective* or *hybrid*; their variances are determined by correlators and parity-type observables. (**?**)

Local observables: baseline diagnostics (often entanglement-blind)

Spin local:

$$\left\langle \sigma_\alpha^{(A)} \right\rangle, \qquad \left\langle \sigma_\alpha^{(B)} \right\rangle, \qquad \alpha \in \{x, y, z\}.$$

Mode local:

$$\left\langle \tau_\mu^{(A)} \right\rangle, \qquad \left\langle \tau_\mu^{(B)} \right\rangle, \qquad \mu \in \{x, y, z\}.$$

What they probe: direct coupling of the parameter to one subsystem, calibration (which qubit couples?), drift monitoring.

What they miss: For many entangled probes (all Bell states), all one-body marginals vanish:

$$\left\langle \sigma_\alpha^{(A)} \right\rangle = \left\langle \sigma_\alpha^{(B)} \right\rangle = 0 \quad \forall\, \alpha.$$

Hence local observables alone can fail to reveal the mechanism of entanglement-enhanced sensing.

Two-body correlators: the primary "whole-system" observables

Define the spin correlation tensor

$$C^{(s)}_{\alpha\beta} = \left\langle \sigma^{(A)}_{\alpha} \sigma^{(B)}_{\beta} \right\rangle, \qquad \alpha, \beta \in \{x, y, z\}.$$

Similarly for spatial modes:

$$C^{(m)}_{\mu\nu} = \left\langle \tau^{(A)}_{\mu} \tau^{(B)}_{\nu} \right\rangle.$$

Connected correlators remove product contributions:

$$\left\langle \sigma^{(A)}_{\alpha} \sigma^{(B)}_{\beta} \right\rangle_c = \left\langle \sigma^{(A)}_{\alpha} \sigma^{(B)}_{\beta} \right\rangle - \left\langle \sigma^{(A)}_{\alpha} \right\rangle \left\langle \sigma^{(B)}_{\beta} \right\rangle.$$

Metrological content: Second moments of collective generators are fixed by correlators. Example:

$$J_z = \frac{1}{2}(\sigma^{(A)}_z + \sigma^{(B)}_z), \quad J_z^2 = \frac{1}{2}\mathbb{I} + \frac{1}{2}\sigma^{(A)}_z \sigma^{(B)}_z.$$

Therefore $\left\langle \sigma^{(A)}_z \sigma^{(B)}_z \right\rangle$ directly determines $\left\langle J_z^2 \right\rangle$ and helps determine $(\Delta J_z)^2$ and thus QFI for common-mode sensing. (**?**)

Measuring a Pauli string: basis-change principle

Most observables reduce to Pauli strings. Measure $\langle P \rangle$ where

$$P = \bigotimes_{k=1}^{n} P_k, \quad P_k \in \{X, Y, Z\}.$$

Procedure:

1. Apply a local unitary U_k to map P_k to Z:

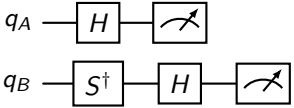
$$X = HZH, \qquad Y = (HS^\dagger)Z(SH).$$

2. Measure all qubits in Z.
3. Multiply outcomes to estimate $\langle P \rangle$.

This is the standard measurement primitive used in VQE, tomography, and correlator estimation. (**?**)

Circuit sketch: measuring $\left\langle \sigma_x^{(A)} \sigma_y^{(B)} \right\rangle$ (example)

Goal: estimate $\langle X_A Y_B \rangle$.



If measurement outcomes are mapped to ± 1 values m_A, m_B, then for ν shots:

$$\widehat{\langle X_A Y_B \rangle} = \frac{1}{\nu} \sum_{k=1}^{\nu} m_A^{(k)} m_B^{(k)}.$$

Correlator estimation from counts (explicit formula)

For two-qubit dichotomic outcomes $(m_A, m_B) \in \{\pm 1\}^2$, counts $N_{++}, N_{+-}, N_{-+}, N_{--}$ yield

$$\widehat{E} = \widehat{\langle A \otimes B \rangle} = \frac{N_{++} + N_{--} - N_{+-} - N_{-+}}{N_{\mathrm{tot}}}.$$

Shot scaling (binomial / bounded outcomes):

$$\mathrm{Var}(\widehat{E}) \approx \frac{1 - E^2}{\nu} \leq \frac{1}{\nu},$$

so additive error ϵ typically costs $\nu = \mathcal{O}(\epsilon^{-2})$ shots per setting.

Importance: measurement setting count $\times\ \epsilon^{-2}$ dominates budgets, motivating direct measurement (few settings) over full tomography when possible.

Collective spin observables: J and J^2

Define collective spin operators:

$$J_\alpha = \frac{1}{2}(\sigma_\alpha^{(A)} + \sigma_\alpha^{(B)}), \qquad \alpha \in \{x, y, z\}.$$

Second moments:

$$J_\alpha^2 = \frac{1}{2}\mathbb{I} + \frac{1}{2}\sigma_\alpha^{(A)}\sigma_\alpha^{(B)}.$$

Thus measuring $\left\langle \sigma_\alpha^{(A)}\sigma_\alpha^{(B)} \right\rangle$ gives $\langle J_\alpha^2 \rangle$.

For pure probes and unitary encoding generated by J_z:

$$F_Q = 4(\Delta J_z)^2.$$

Hence collective fluctuations are directly metrological. (**??**)

Mode collective operators are analogous:

$$K_\mu = \frac{1}{2}(\tau_\mu^{(A)} + \tau_\mu^{(B)}).$$

Total spin $\mathbf{J}^2$ and singlet/triplet discrimination

Define

$$\mathbf{J}^2 = J_x^2 + J_y^2 + J_z^2 = \frac{3}{2}\mathbb{I} + \frac{1}{2}\left(\sigma_x^{(A)}\sigma_x^{(B)} + \sigma_y^{(A)}\sigma_y^{(B)} + \sigma_z^{(A)}\sigma_z^{(B)}\right).$$

Eigenvalues:

$$\mathbf{J}^2 \left|j, m\right\rangle = j(j+1)\left|j, m\right\rangle \quad \Rightarrow \quad j = 1 : \mathbf{J}^2 = 2, \;\; j = 0 : \mathbf{J}^2 = 0.$$

Measurement: estimate XX, YY, ZZ correlators and combine linearly to obtain $\left\langle \mathbf{J}^2 \right\rangle$.

Sensing relevance: singlet ($j = 0$) is immune to common-mode generators like J_z, while triplets respond. This is important for common-mode rejection vs gradient sensing.

Parity: definition and rotated parity readout

Spin z-parity:

$$\Pi_z^{(s)} = \sigma_z^{(A)}\sigma_z^{(B)} \in \{+1, -1\}.$$

Mode parity similarly:

$$\Pi_z^{(m)} = \tau_z^{(A)}\tau_z^{(B)}.$$

Rotated parity: apply a (possibly collective) rotation $R(\theta)$ then measure z-parity:

$$\Pi^{(s)}(\theta) = R^\dagger(\theta)\,\Pi_z^{(s)}\,R(\theta).$$

For example, measuring X-parity is just Hadamards before Z readout:

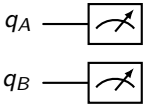
$$X\text{-parity } = \sigma_x^{(A)}\sigma_x^{(B)} = (H \otimes H)\,\Pi_z^{(s)}\,(H \otimes H).$$

Parity measurements are widely used as high-contrast interferometric readout observables, often approaching optimality for certain nonclassical probes. (**??**)

Parity measurement I: direct (no ancilla)

Measure both qubits in Z and post-process:

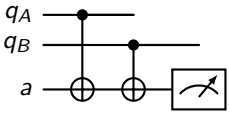
$$\Pi_z^{(s)} = m_A m_B, \qquad m_A, m_B \in \{\pm 1\}.$$



Rotated parity: insert basis changes (e.g., H or $S^\dagger H$ gates) before measurement. This becomes a Pauli-string measurement of XX, YY, etc.

Parity measurement II: ancilla-based parity mapping

If hardware favors reading out one qubit, map parity to an ancilla a:



Initialize a in $|0\rangle$: the ancilla measurement outcome satisfies $m_a = m_A m_B$, i.e. it returns parity. For X- or Y-parity, apply basis changes on q_A, q_B before the CNOT ladder.

Bell/CHSH operator (built from correlators)

Choose dichotomic observables (outcomes ± 1)

$$A_0, A_1 \text{ on } A; \qquad B_0, B_1 \text{ on } B.$$

CHSH Bell operator:

$$\mathcal{B}_{\mathrm{CHSH}} = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1).$$

Define correlators $E_{xy} = \langle A_x \otimes B_y \rangle$, then

$$S = E_{00} + E_{01} + E_{10} - E_{11} = \langle \mathcal{B}_{\mathrm{CHSH}} \rangle .$$

Local hidden-variable theories imply $|S| \leq 2$; quantum mechanics allows $|S| \leq 2\sqrt{2}$. (**?**)

Sensing relevance: CHSH is not a phase estimator per se, but it is a compact diagnostic of nonlocal correlation quality (useful under noise/SPAM and for validating entanglement resources used in sensing).

Canonical CHSH measurement settings and basis rotations

A standard choice that maximally violates CHSH on $|\Phi^+\rangle$ is

$$A_0 = Z, \quad A_1 = X, \qquad B_0 = \frac{Z+X}{\sqrt{2}}, \quad B_1 = \frac{Z-X}{\sqrt{2}}.$$

Realize B_0, B_1 by single-qubit rotations on B:

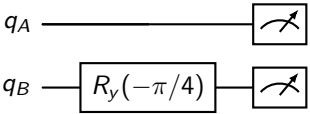
$$B_0 = R_y(-\tfrac{\pi}{4})\, Z\, R_y(\tfrac{\pi}{4}), \qquad B_1 = R_y(\tfrac{\pi}{4})\, Z\, R_y(-\tfrac{\pi}{4}).$$

Thus each correlator is a $Z \otimes Z$ measurement in an appropriate rotated basis.

Computing S requires four settings $(E_{00}, E_{01}, E_{10}, E_{11})$.

Circuit sketch: measuring one CHSH correlator E_{00}

Example: $E_{00} = \langle A_0 \otimes B_0 \rangle = \left\langle Z \otimes (Z + X)/\sqrt{2} \right\rangle$.



Then compute $\widehat{E}_{00}$ from the ± 1 outcomes and repeat for the other three settings.

Hybrid spin–mode observables (essential for position-dependent sensing)

Define local hybrid operators:

$$O_A^{(sz)} = \sigma_z^{(A)} \tau_z^{(A)}, \qquad O_B^{(sz)} = \sigma_z^{(B)} \tau_z^{(B)}.$$

Two-system hybrid correlators:

$$C_{\alpha\mu,\beta\nu}^{(sm)} = \left\langle (\sigma_\alpha^{(A)} \tau_\mu^{(A)})(\sigma_\beta^{(B)} \tau_\nu^{(B)}) \right\rangle .$$

Hybrid parity-like string:

$$\Pi_{zz}^{(sm)} = (\sigma_z^{(A)} \tau_z^{(A)})(\sigma_z^{(B)} \tau_z^{(B)}).$$

Metrological necessity: If encoding generator is $G_{sz} = \frac{1}{2}(\sigma_z^{(A)} \tau_z^{(A)} + \sigma_z^{(B)} \tau_z^{(B)})$, then

$$F_Q(\text{pure}) = 4(\Delta G_{sz})^2$$

depends on hybrid correlators and cannot, in general, be inferred from spin-only measurements.

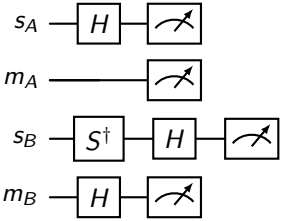
Measurement: treat spin and mode qubits as four physical qubits and measure the corresponding Pauli string via basis changes and Z readout.

Circuit sketch: measuring a hybrid Pauli string (example)

Example hybrid observable:

$$P = (\sigma_x^{(A)}\tau_z^{(A)})(\sigma_y^{(B)}\tau_x^{(B)}) \quad \text{(a Pauli string on four qubits).}$$

Assume qubit ordering: (s_A, m_A, s_B, m_B).



Post-process by multiplying all four ± 1 outcomes to estimate $\langle P \rangle$.

Worked example 1: Bell states (spin) and correlator signatures

Bell states:

$$|\Phi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \qquad |\Psi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle).$$

All one-body marginals vanish: $\left\langle \sigma_\alpha^{(A)} \right\rangle = \left\langle \sigma_\alpha^{(B)} \right\rangle = 0.$

Key correlators (spin):

$$\langle XX \rangle\,,\ \langle YY \rangle\,,\ \langle ZZ \rangle\,.$$

State	$\left\langle \sigma_x^{(A)} \sigma_x^{(B)} \right\rangle$	$\left\langle \sigma_y^{(A)} \sigma_y^{(B)} \right\rangle$	$\left\langle \sigma_z^{(A)} \sigma_z^{(B)} \right\rangle$
$\lvert\Phi^+\rangle$	$+1$	-1	$+1$
$\lvert\Phi^-\rangle$	-1	$+1$	$+1$
$\lvert\Psi^+\rangle$	$+1$	$+1$	-1
$\lvert\Psi^-\rangle$ (singlet)	-1	-1	-1

Thus Bell states are best identified and monitored via correlators/parity, not local means.

Worked example 2: explicit QFI for common-mode vs differential encoding (spin)

Generators:

$$G_{\mathrm{com}}^{(s)} = \frac{1}{2}(\sigma_z^{(A)} + \sigma_z^{(B)}), \qquad G_{\mathrm{diff}}^{(s)} = \frac{1}{2}(\sigma_z^{(A)} - \sigma_z^{(B)}).$$

For pure probes, $F_Q = 4(\Delta G)^2$.

Probe	F_Q for $G_{\mathrm{com}}^{(s)}$ (common)	F_Q for $G_{\mathrm{diff}}^{(s)}$ (diff)
$\lvert+\rangle\lvert+\rangle$	2	2
$\lvert\Phi^+\rangle$	4	0
$\lvert\Psi^+\rangle$	0	4
$\lvert\Psi^-\rangle$ (singlet)	0	4

Interpretation:

- $\lvert\Phi^+\rangle$ is an optimal two-qubit probe for common-mode Z phase (Heisenberg-like factor 2 in phase).
- $\lvert\Psi^+\rangle$ or $\lvert\Psi^-\rangle$ are optimal for gradients/differentials.

Worked example 3: singlet/triplet sensors (symmetry and immunity)

Triplet states:

$$|T_+\rangle = |00\rangle\,, \quad |T_0\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \quad |T_-\rangle = |11\rangle\,,$$

Singlet:

$$|S\rangle = |\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle).$$

Total spin:

$$\mathbf{J}^2\,|T_m\rangle = 2\,|T_m\rangle\,, \qquad \mathbf{J}^2\,|S\rangle = 0.$$

Common-mode field ($G^{(s)}_{\mathrm{com}} \propto J_z$):

$$G^{(s)}_{\mathrm{com}}\,|S\rangle = 0 \quad \Rightarrow \quad (\Delta\, G^{(s)}_{\mathrm{com}})^2 = 0 \Rightarrow F_Q = 0.$$

Thus the singlet is a **common-mode rejector**.

Differential field ($G^{(s)}_{\mathrm{diff}}$): acts within the $\{|01\rangle\,, |10\rangle\}$ subspace and generates relative phases; $|T_0\rangle$ and $|S\rangle$ are gradient-sensitive probes.

Worked example 4: spatial (mode) Bell state as a gradiometer

Mode basis $|L\rangle, |R\rangle$ with $\tau_z |L\rangle = +|L\rangle$ and $\tau_z |R\rangle = -|R\rangle$.
Spatial Bell-like state:

$$|\Psi_m^+\rangle = \frac{1}{\sqrt{2}}(|L_A R_B\rangle + |R_A L_B\rangle).$$

Then (mode correlators):

$$\left\langle \tau_z^{(A)} \tau_z^{(B)} \right\rangle = -1, \qquad \left\langle \tau_x^{(A)} \tau_x^{(B)} \right\rangle = +1, \qquad \left\langle \tau_y^{(A)} \tau_y^{(B)} \right\rangle = +1.$$

Mode generators:

$$G_{\mathrm{com}}^{(m)} = \frac{1}{2}(\tau_z^{(A)} + \tau_z^{(B)}), \qquad G_{\mathrm{diff}}^{(m)} = \frac{1}{2}(\tau_z^{(A)} - \tau_z^{(B)}).$$

For $|\Psi_m^+\rangle$:

$$F_Q(|\Psi_m^+\rangle, G_{\mathrm{com}}^{(m)}) = 0, \qquad F_Q(|\Psi_m^+\rangle, G_{\mathrm{diff}}^{(m)}) = 4.$$

So spatial entanglement naturally realizes a **two-mode gradiometer**.

Worked example 5: rotated parity fringes for $|\Phi^+\rangle$ under common-mode encoding

Prepare $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ and encode via $U(\phi) = e^{-i\phi G^{(s)}_{\mathrm{com}}}$. Up to a global phase:

$$|\psi(\phi)\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + e^{i2\phi}\,|11\rangle\big).$$

Then

$$\left\langle \sigma_x^{(A)}\sigma_x^{(B)} \right\rangle = \cos(2\phi), \qquad \left\langle \sigma_y^{(A)}\sigma_y^{(B)} \right\rangle = -\cos(2\phi).$$

Parity-as-correlator readout: Measure XX by applying Hadamards and measuring ZZ parity:

$$\langle XX\rangle = \left\langle (H\otimes H)\,\Pi_z^{(s)}\,(H\otimes H) \right\rangle.$$

This yields fringes with doubled phase frequency (slope enhancement), consistent with $F_Q = 4$.

This is a prototypical demonstration of how a whole-system observable (correlator/parity) accesses entanglement-enhanced phase encoding.

Comparison table: observables vs sensing targets and metrology

Observable	**What it measures**	**Common vs diff sensitivity**	**Entanglement / QFI relevance**
Local $\sigma_\alpha^{(A)}$ (or $\tau_\mu^{(A)}$)	Single-subsystem polarization/occupation	Sensitive to local couplings; can miss global encoding	Often blind to entanglement; insufficient to certify metrological gain
Two-body $\sigma_\alpha^{(A)}\sigma_\beta^{(B)}$	Correlation tensor component	Separates $\lvert\Phi\rangle$-type (common) vs $\lvert\Psi\rangle$-type (diff) tasks	Determines second moments of J_α; feeds $(\Delta G)^2$ and thus QFI
Collective J_α, J_α^2	Global magnetization + fluctuations	Directly aligned with common-mode generators ($G_{\mathrm{com}}^{(s)}$)	For pure probes: $F_Q = 4(\Delta J_\alpha)^2$
Parity $\Pi^{(s)}(\theta)$	Even/odd outcomes after readout rotation	High-contrast readout for phase fringes; works for both common/diff with correct rotation	Often near-optimal (state-dependent); relates to Pauli-string expectations
Bell/CHSH S	Scalar summary of multiple correlators	Not a phase estimator but diagnoses nonlocal correlations under noise	Tracks entanglement quality; useful validation tool for sensing resources
Hybrid $\sigma_z^{(A)}\tau_z^{(A)}$, etc.	Spin conditioned on mode; joint spin–mode correlations	Essential if encoding is mode-dependent or hybrid (G_{com})	Governs $(\Delta G)^2$ for hybrid generators; can reveal hybrid entanglement useful for sensing

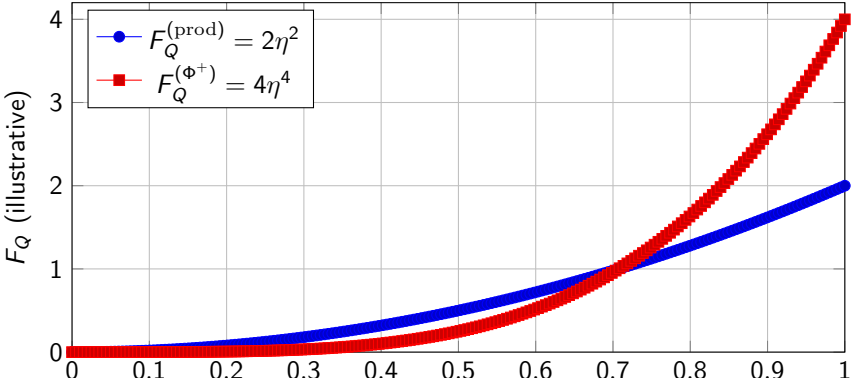
Noise-limited metrology: QFI under dephasing (two-qubit illustration)

Independent dephasing is a canonical noise model for sensors (e.g., frequency standards). It can destroy entanglement advantage beyond a crossover time. (**???**)

Let $\eta = e^{-\Gamma t}$ denote single-qubit coherence. A simple illustrative scaling for common-mode sensing:

$$F_Q^{(\mathrm{prod})} \approx 2\eta^2, \qquad F_Q^{(\Phi^+)} \approx 4\eta^4.$$

Entangled probes can be best near $\eta \simeq 1$, but degrade faster under uncorrelated dephasing. (**?**)



Dephasing channel model (Kraus form) and impact on observables

A common single-qubit dephasing (phase-flip) channel:

$$\mathcal{E}_p(\rho) = (1-p)\rho + p\, Z\rho Z,$$

with Kraus operators $K_0 = \sqrt{1-p}\,\mathbb{I}$, $K_1 = \sqrt{p}\,Z$.

Two-qubit independent dephasing:

$$\rho \mapsto (\mathcal{E}_p \otimes \mathcal{E}_p)(\rho).$$

Observable effects:

- Coherences (and thus XX, YY, rotated parity fringes, CHSH) decay with p.
- ZZ parity may remain robust if populations are unchanged.
- QFI and FI decrease; entanglement advantage can vanish. (**??**)

Readout/SPAM errors and mitigation (platform unspecified)

Readout (SPAM) errors bias correlators and parity because they depend on products of outcomes.

Typical mitigation approaches (platform-dependent):

- Calibrate a **confusion matrix** M for outcome flips and invert/regularize classically.
- Use measurement symmetrization / randomized compiling to average coherent readout bias.
- Prefer a **minimal measurement program** tailored to the sensing generator to reduce the number of settings and accumulation of bias.

Because the physical platform is unspecified, we do not assume a particular mitigation pipeline; instead, we emphasize robust measurement choices (parity/correlators) plus calibration.

Tomography vs direct measurement (settings and shot cost)

Two-qubit state tomography requires up to 15 nontrivial Pauli expectations (excluding $\mathbb{I} \otimes \mathbb{I}$), usually grouped into a few local bases.

For sensing: direct measurement is usually better:

- If theory predicts a generator G and an (approx.) optimal measurement, measure only what you need: $\langle P \rangle$ for a small set of Pauli strings P that determine F_C or validate preparation.
- Example common-mode sensing with $|\Phi^+\rangle$: measure XX (via rotated parity) plus ZZ (parity) and optionally CHSH for entanglement validation.

Shot scaling: additive error ϵ per setting typically costs $\nu = \mathcal{O}(\epsilon^{-2})$ shots. Total cost scales roughly as $(\#\text{settings})/\epsilon^2$.

HEP mapping I: spin density matrices and correlation tensors in $t\bar{t}$

In HEP, two-particle spin states are often described by a 4×4 density matrix decomposed as

$$\rho = \frac{1}{4}\left(\mathbb{I} \otimes \mathbb{I} + \sum_i B_i\, \sigma_i \otimes \mathbb{I} + \sum_j \bar{B}_j\, \mathbb{I} \otimes \sigma_j + \sum_{ij} C_{ij}\, \sigma_i \otimes \sigma_j\right).$$

Here C_{ij} is the **spin correlation matrix**.

Direct mapping to sensor correlators:

$$C_{ij} \quad \longleftrightarrow \quad \left\langle \sigma_i^{(A)} \sigma_j^{(B)} \right\rangle.$$

In top-pair physics, C_{ij} is accessed via angular distributions of decay products, effectively implementing local spin measurements. (**?**)

Thus the same correlators used in quantum sensing diagnostics are canonical in HEP spin-correlation measurements.

HEP mapping II: Bell tests with neutral mesons (kaons, B mesons)

Neutral meson pairs (e.g., $K^0\bar{K}^0$) form entangled "quasi-spin" systems. (**?**)

Operator mapping idea: Choose a two-level basis, e.g.

$$|K^0\rangle \equiv |0\rangle\,, \qquad |\bar{K}^0\rangle \equiv |1\rangle\,.$$

Then strangeness measurement corresponds to a Pauli-Z-type observable on quasi-spin:

$$S \sim \sigma_z.$$

Other measurement choices (CP eigenstates, different decay channels, or time-tagged measurements) correspond to rotated Pauli axes, enabling Bell-type inequalities.

Analogy to sensing: multiple correlator settings (CHSH) are measured in different bases/times, similar to rotating measurement bases in two-qubit sensors.

HEP mapping III: relativistic detector models (Unruh–DeWitt) and harvesting

In relativistic QFT, localized detectors are often modeled as two-level systems interacting with a quantum field. One studies two-detector correlations and entanglement "harvesting" from the vacuum. (**??**)

Correlation diagnostics are precisely two-qubit observables:

$$\left\langle \sigma_\alpha^{(A)} \sigma_\beta^{(B)} \right\rangle, \qquad \left\langle \sigma_\alpha^{(A)} \right\rangle, \qquad \langle \mathcal{B}_{\mathrm{CHSH}} \rangle .$$

Parameters like detector separation, switching functions, and acceleration play roles analogous to sensing parameters ϕ in $\rho(\phi)$.

Thus, the sensor observable toolkit applies directly to relativistic quantum information experiments and theory.

HEP mapping IV: neutrino/flavor qubits and oscillation interferometry

Two-flavor neutrino oscillations can be mapped to a qubit:

$$|\nu_e\rangle \equiv |0\rangle\,, \qquad |\nu_\mu\rangle \equiv |1\rangle\,,$$

and flavor measurement corresponds to a Pauli-Z measurement in the flavor basis:

$$M_{\text{flavor}} \sim \sigma_z.$$

The oscillation Hamiltonian is a linear combination of Pauli matrices, giving Bloch-sphere rotations.

Multi-flavor oscillations can be viewed as multi-mode entanglement of a single particle across flavor modes, with quantum-correlation and Bell-type analyses explored in the literature. (**?**)

Main structural link: **generator + measurement** architecture shared between oscillation interferometry and quantum sensing.

Table: mapping sensor observables to HEP analogues

Sensor observable	**Meaning in sensing**	**HEP analogue / mapping**
$\left\langle \sigma_i^{(A)} \sigma_j^{(B)} \right\rangle$	Spin correlation tensor component	Spin correlation matrix C_{ij} in $t\bar{t}$ and other pair production; accessed via decay angles (**?**)
$\langle \mathcal{B}_{\mathrm{CHSH}} \rangle$	Bell nonlocality diagnostic; entanglement quality	Bell tests with neutral mesons (kaons, B mesons), rotated bases/time-tagged correlators (**?**)
Parity $\left\langle \Pi_z^{(s)} \right\rangle$ and rotated parity	High-contrast global readout; interferometric fringes	Parity-/CP-/flavor-type correlators in unstable systems; basis rotations correspond to different decay channels/time evolution (**?**)
Generator variance $(\Delta G)^2$	Pure-state QFI via $F_Q = 4(\Delta G)^2$	Oscillation / phase sensitivity to mass splittings and phases (neutrinos, mesons); detector-parameter sensitivity in QFT models (**??**)
Hybrid spin–mode strings	Position-conditioned response	Two-detector spatial separation effects and local couplings in relativistic detector models (**??**)

Python simulation goals (what the code demonstrates)

We provide a small NumPy-only script that:

- prepares product and Bell probes (spin-only, for clarity; extension to mode/hybrid is straightforward),
- encodes a parameter via common-mode and differential generators,
- applies an independent dephasing channel (mixed states),
- computes QFI using the spectral formula (reduces to $4(\Delta G)^2$ for pure states),
- computes classical FI for specific measurement choices (e.g., XX correlator readout for Bell probes).

The code is self-contained (no Qiskit needed). Save as `simulate_two_qubit_sensor.py` and run:

```
python simulate_two_qubit_sensor.py
```

Note: This slide deck does not execute Python; it includes code listings for you to run locally.

Python (1/3): states, generators, unitary encoding, dephasing channel

```
import numpy as np

# ---------- basic matrices ----------
I = np.eye(2, dtype=complex)
X = np.array([[0, 1],[1, 0]], dtype=complex)
Y = np.array([[0,-1j],[1j,0]], dtype=complex)
Z = np.array([[1, 0],[0,-1]], dtype=complex)

def kron(*ops):
    out = np.array([[1]], dtype=complex)
    for op in ops:
        out = np.kron(out, op)
    return out

# Two-qubit Paulis (spin only; extend similarly for 4-qubit spin+mode)
ZA, ZB = kron(Z,I), kron(I,Z)
XA, XB = kron(X,I), kron(I,X)

# ---------- probe states ----------
zero = np.array([[1],[0]], dtype=complex)
```

Python (2/3): QFI (spectral formula) and FI for a chosen measurement

```
def qfi_unitary(rho, G, tol=1e-12):
    """
    QFI for unitary encoding rho(phi)=e^{-i phi G} rho e^{+i phi G}.
    Uses the spectral formula:
    F_Q = 2 sum_{i,j} ( (li-lj)^2/(li+lj) ) |<i|G|j>|^2 , li+lj>0
    """
    lam, U = np.linalg.eigh(rho)
    # sort descending (optional)
    idx = np.argsort(lam)[::-1]
    lam = lam[idx]
    U = U[:, idx]

    G_ij = U.conj().T @ G @ U
    F = 0.0
    for i in range(len(lam)):
        for j in range(len(lam)):
            denom = lam[i] + lam[j]
            if denom > tol:
                num = (lam[i] - lam[j])**2
                F += 2.0 * (num/denom) * (abs(G_ij[i,j])**2)
```

Python (3/3): run a small comparison table and print results

```
def demo():
    phis = [0.2, 0.5] # sample phases
    ps = [0.0, 0.05, 0.15] # dephasing strengths

    for p in ps:
        print(f"\n=== Dephasing p = {p:.2f} ===")
        for phi in phis:
            # QFI (spectral) for common-mode generator
            qfi_prod = qfi_unitary(dephase_two_qubit(rho_prod, p), G_com)
            qfi_phi = qfi_unitary(dephase_two_qubit(rho_phi, p), G_com)

            # Classical FI for XX measurement (optimal for Bell common-mode case)
            fi_prod = FI_for_measurement(rho_prod, G_com, phi, dephase_p=p)
            fi_phi = FI_for_measurement(rho_phi, G_com, phi, dephase_p=p)

            print(f"phi={phi: .2f} | QFI(prod)={qfi_prod: .4f} QFI(Phi+)={qfi_phi: .4f} "
                  f"| FI_XX(prod)={fi_prod: .4f} FI_XX(Phi+)={fi_phi: .4f}")

    # Also show task-dependence: QFI of Phi+ under differential generator should be ~0 (
        ideal case)
```

Optional Mermaid flowchart (external rendering only)

Beamer does not natively render Mermaid. If your build pipeline supports Mermaid rendering, you can use this source:

```
flowchart TD
  A[Prepare probe rho0] --> B[Encode: U(phi)=exp(-i phi G)]
  B --> C[Choose measurement setting (basis rotations)]
  C --> D[Run shots, collect bitstrings]
  D --> E[Estimate <O> or p(o|phi)]
  E --> F[Compute FI; compare to QFI]
  F --> G[Assess entanglement role; common vs differential sensitivity]
```

Recommended minimal measurement program (actionable checklist)

A minimal yet powerful program depends on the sensing generator G.

Always (baseline and calibration):

- Local Z readout: $\left\langle \sigma_z^{(A)} \right\rangle, \left\langle \sigma_z^{(B)} \right\rangle$ (and if relevant $\left\langle \tau_z^{(A)} \right\rangle, \left\langle \tau_z^{(B)} \right\rangle$).
- Parity ZZ (spin): $\left\langle \Pi_z^{(s)} \right\rangle = \left\langle \sigma_z^{(A)} \sigma_z^{(B)} \right\rangle$ (and mode parity $\left\langle \Pi_z^{(m)} \right\rangle$ if using spatial entanglement).

If sensing common-mode spin phase ($G = G_{\mathrm{com}}^{(s)}$):

- Measure rotated parity / correlator XX via $(H \otimes H)$ then ZZ readout.
- Optionally measure YY (via $S^\dagger H$) to validate Bell coherence.
- Optional validation: CHSH S (4 settings) as an entanglement-quality benchmark.

If sensing differential spin phase ($G = G_{\mathrm{diff}}^{(s)}$):

- Use $|\Psi\rangle$-type probes; measure observables sensitive to relative phase in the $\{|01\rangle, |10\rangle\}$ subspace, typically XX and YY.

If sensing mode or hybrid coupling (G uses τ or $\sigma\tau$):

- Measure the corresponding mode correlators/parity (for τ) and hybrid Pauli strings (for $\sigma\tau$).

Suggested experiments (platform-agnostic prototypes)

Because platform details are unspecified, we phrase experiments in terms of generic controls.

Experiment A: Uniform-field entanglement gain (spin).

- Prepare $|+\rangle\,|+\rangle$ and $|\Phi^+\rangle$.
- Encode ϕ via $U(\phi) = e^{-i\phi G^{(s)}_{\mathrm{com}}}$ (same Z phase on both qubits).
- Read out via XX (Hadamards then ZZ parity). Compare FI vs QFI.

Experiment B: Gradient selectivity (spin).

- Prepare $|\Phi^+\rangle$ vs $|\Psi^+\rangle$.
- Encode gradient phase via $U(\phi) = e^{-i\phi G^{(s)}_{\mathrm{diff}}}$ (opposite Z phases).
- Show $|\Psi^+\rangle$ yields high sensitivity while $|\Phi^+\rangle$ yields near-zero.

Experiment C: Spatial Bell as a gradiometer (mode).

- Prepare $|\Psi^+_m\rangle = (|LR\rangle + |RL\rangle)/\sqrt{2}$ in the mode qubits.
- Apply differential mode phase; read out mode parity/correlators ($\tau_x\tau_x$, $\tau_z\tau_z$).

Experiment D: Hybrid sensing (spin–mode).

How to compile and run (copy-paste instructions)

To compile the slides:

1. Copy this entire `.tex` content into `two_qubit_sensor_observables.tex`.
2. Compile with:

   ```
   pdflatex two_qubit_sensor_observables.tex
   ```

   Run twice to resolve citations.
3. Alternatively, use `xelatex` if you prefer.

To run the Python simulation:

1. Copy the code listings into `simulate_two_qubit_sensor.py`.
2. Run:

   ```
   python simulate_two_qubit_sensor.py
   ```

Sharing/downloading options:

- Easiest: copy-paste into a local file (no hosting required).
- Alternatively: upload to Overleaf (one-click compilation) or host on GitHub and share a link.

References (primary reviews and key papers)

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